SUSHMITHA SUDHARSAN

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SUMMARY

AI/ML Engineer with an MS in Data Analytics Engineering, specializing in production-grade MLOps and applied GenAI. Built end-to-end ML pipelines with automated CI/CD, retraining, and drift monitoring, sustaining sub-200ms latency and reducing data errors by 80%. Proficient in PyTorch, MLflow, Airflow, Docker, and GCP. First-author IEEE publication in applied deep learning.

EDUCATION

Northeastern University Apr 2026
Master of Science in Data Analytics Engineering, GPA: 3.92/4.0 *Boston, MA*
- **Coursework:** Data Mining, Generative AI, MLOps, Machine Learning, Deep Learning for AI, Data Visualization

SRM Institute of Science and Technology May 2024
Bachelor of Technology in Computer Science (AI & ML Specialization) *Chennai, India*

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, C++, Pandas, NumPy, Apache Airflow, ETL Pipelines, API Integration
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM
GenAI & NLP: LLMs, RAG, LangChain, Multi-Agent Systems, NLP, Prompt Engineering
MLOps & Deployment: MLflow, Docker, CI/CD, FastAPI, Model Monitoring, Prometheus, DVC, Great Expectations, PyTest
Cloud & Vision: GCP, Cloud Run, HuggingFace Spaces, YOLOv8, Object Detection, OpenCV, CLIP Embeddings

PROJECTS

MLOps Financial Forecasting Platform [GitHub] | *PyTorch, XGBoost, Airflow, MLflow, GCP* Sep – Dec 2025
- Built a forecasting platform with a 6-member team to predict how 100 companies' stocks react to crises (2008, COVID), automating PyTorch and XGBoost pipelines in Airflow and validating 2.5M+ records with Great Expectations
- Owned CI/CD for automated retraining, GCP deployment, and PyTest testing; tracked 50+ MLflow experiments and monitored drift with Prometheus/Grafana at under 200ms p99

GenAI Sports Analysis Chatbot [GitHub] | *LangChain, FAISS, GPT-4o, Gradio* Mar 2026
- Built a chatbot that answers World Cup stats questions across 49,000+ matches and 47,000+ goals, designing a ReAct agent that connects GPT-4o, embeddings, FAISS, and 5 tools
- Shipped in 48 hours with evidence-backed answers (confidence scores, stated limitations) and a match-outcome predictor

VisionScript: Multimodal Image Captioning [GitHub] | *CLIP, BLIP, GPT-2, PyTorch* Jan – Apr 2026
- Built a tone-adjustable image captioning pipeline (e.g., child-friendly) with a 4-member team, linking CLIP to GPT-2 through a custom projection layer fine-tuned on 1,000+ Flickr30k pairs
- Tested decoding strategies to reduce hallucinated captions; deployed a live demo on HuggingFace Spaces

ICU Mortality & Length-of-Stay Prediction (MIMIC-IV) [GitHub] | *TensorFlow, XGBoost, Random Forest* Jan – Apr 2025
- Trained a Random Forest, XGBoost, and neural network ensemble on 94,000+ ICU admissions with 45+ temporal features
- Reached 0.945 AUC for mortality and 0.88 AUC for length-of-stay prediction under HIPAA-compliant preprocessing

EXPERIENCE

Data Analyst Intern Jun 2023 – Sep 2023
Whirl Data Inc. *Chennai, India*
- Analyzed 10,000+ dashcam frames to find 15+ factors (lighting, placement, plate condition, speed) hurting plate capture
- Built a Python/OpenCV pipeline scoring brightness, Laplacian sharpness, and contrast into 3 quality tiers; ~10% unusable
- Extracted plate text with Tesseract OCR; proposed duplicate and anomaly-frame removal for plate-detection training

PUBLICATIONS & ACHIEVEMENTS

- **IEEE Publication (First Author) [IEEE Xplore]:** Developed an RNN/LSTM model to track classroom engagement and flag student fatigue, achieving 96% accuracy with 2–3 sec real-time latency across 30 participants
- **Finalist, Data Mining Hackathon:** Led a team building an ML pipeline in 48 hours, achieving a 0.89 F1-score (2nd of 25 teams)

CERTIFICATIONS

- Azure Fundamentals (AZ-900) | Azure AI Fundamentals (AI-901) | Power BI | AWS Generative AI | AWS Prompt Engineering